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Healthcare AI researcher with a rigorous mathematical foundation (B.S. Mathematics, Korea University) focused on two problems where errors carry clinical consequences: automated coronary vessel analysis and LLM reliability in structured medical knowledge. 25 SCI/SCIE publications (6 co-first, 1 corresponding, 404 citations, h-index 12) within two years. Seeking a PhD research environment to pursue these problems at depth — Fall 2027.

EDUCATION

M.S. — Computational Health & Biomedical Data Science (School of Pharmacy, Kyung Hee University)
Mar. 2023 – Feb. 2025

Thesis: “Development of A Machine Learning-Based Prediction Algorithm for Suicidal Ideation in Adolescents and Independent Validation Across Three Worldwide Cohorts”

Advisor: Professor Sang Youl Rhee & DongKeon Yon

B.S. Korea University, South Korea

Mar. 2016 – Feb. 2023

Major: Mathematics

FELLOWSHIPS & VISITING APPOINTMENTS

Ministry of Science and ICT (MSIT/IITP) National Fellowship

University of Toronto, Dept. of Mechanical & Industrial Engineering | Jan 2024 – Jul 2024

Fully funded government fellowship for applied AI research; collaborated with MIT research team on LLM hallucination mitigation in scientific Q&A.

PROFESSIONAL & RESEARCH EXPERIENCE

PUBLICATIONS

*Co-first author; †Corresponding author.

1. Son Y, Hong S, Yim Y, Kim S, Lee H, Lee K, Kim HJ, Jo H, Park J, Oh J, Lee S, Lee H, Nehs CJ, Smith L, Yon DK, Kang J. Global prevalence of cannabis and amphetamine/methamphetamine use among adolescents in 47 countries: a population-based study from WHO database. *World J Pediatr.* 2025 Mar; 21(3):291-305. (IF: 4.4)
2. Kim S, Kim H, Kim S, Lee H, Hammoodi A, Choi Y, Kim HJ, Smith L, Kim MS, Fond G, Boyer L, Baik SW, Lee H, Park J, Kwon R, Woo S, Yon DK. Machine Learning-Based Prediction of Substance Use in Adolescents in Three Independent Worldwide Cohorts: Algorithm Development and Validation Study. *J Med Internet Res.* 2025 Feb 24; 27:e62805. (IF: 7.4)

3. Kim S, Park J, Son Y, Lee H, Woo S, Lee M, Lee H, Sang H, Yon DK, Rhee SY. Development and Validation of a Machine Learning Algorithm for Predicting Diabetes Retinopathy in Patients With Type 2 Diabetes: Algorithm Development Study. *JMIR Med Inform.* 2025 Feb 7; 13:e58107. (IF: 7.4)
4. Yim Y, Lee JE, Son Y, Kim S, Lee H, Lee S, Jang W, Cho H, Lee H, Lee K, Jo H, Park J, Oh J, Jacob L, Kang J, Sang H, Rhee SY, Yon DK. Long-term trends in the prevalence of cardiovascular-kidney-metabolic syndrome in South Korea, 2011-2021: a representative longitudinal serial study. *Lancet Reg Health West Pac.* 2025 Jan 22;55:101474. (IF: 11.1)
5. Sang H, Park J, Kim S, Lee M, Lee H, Lee SH, Yon DK, Rhee SY. Author Correction: Predictive model for abdominal liposuction volume in patients with obesity using machine learning in a longitudinal multi-center study in Korea. *Sci Rep.* 2025 Jan 23; 15(1):2928. (IF: 3.8)
6. Hong S, Son Y, Lee M, Park J, Lee H, Lee H, **Lee H**, Kim H, Dragioti E, Fond G, Boyer L, López Sánchez GF, Tully MA, Rahmati M, Woo S, Yon DK, Smith L. Temporal changes in smoking prevalence among adolescents across 23 countries. *Acta Paediatr.* 2025 Jun;114(6):1189-1201.
7. Lee K, Park J, Lee M, **Lee H**, Son Y, Kim H, Kang J, Choi Y, Rhee SY, Rahmati M, Koyanagi A, Smith L, López Sánchez GF, Dragioti E, Woo S, Yon DK. National trends in thyroid disease and COVID-19 pandemic-related factors, 1998-2021: A nationwide representative study in South Korea. *Heliyon.* 2024 Oct 15;10(20):e39401.
8. Sang H*, **Lee H***, Park J, Kim S, Woo HG, Koyanagi A, Smith L, Lee S, Hwang YC, Park TS, Lim H, Yon DK[¶], Rhee SY[¶]. Machine Learning-Based Prediction of Neurodegenerative Disease in Patients With Type 2 Diabetes by Derivation and Validation in 2 Independent Korean Cohorts: Model Development and Validation Study. *J Med Internet Res.* 2024 Oct 3;26:e56922.
*Co-first author.
9. Sang H*, **Lee H***, Lee M, Park J, Kim S, Woo HG, Rahmati M, Koyanagi A, Smith L, Lee S, Hwang YC, Park TS, Lim H, Yon DK[¶], Rhee SY[¶]. Prediction model for cardiovascular disease in patients with diabetes using machine learning derived and validated in two independent Korean cohorts. *Sci Rep.* 2024 Jun 28;14(1):14966.
*Co-first author.
10. Park S*, Yim Y*, Lee M, **Lee H**, Park J, Lee JH, Woo S, Kim T, Kang J, Smith L, López Sánchez GF, Dragioti E, Rahmati M, Fond G, Boyer L, Shin JI, Son Y, Yon DK[¶], Cortese S[¶]. Longitudinal trends in depression, suicidal ideation, and suicide attempts by family structure in South Korean adolescents, 2009-2022: A nationally representative serial study. *Asian J Psychiatr.* 2024 Aug;98:104122.
11. Kim H*, Kyung S*, Park J*, **Lee H**, Lee M, Smith L, Rahmati M, Shin JY, Kang J, Jacob L, Papadopoulos NG, Rhee SY, Lee J, Kim HJ[¶], Lee H[¶], Yon DK[¶]. Risks of cutaneous immune-related adverse events in long COVID: Multinational cohort studies in South Korea, Japan, and the UK. *J Med Virol.* 2024 Jun;96(6):e29740.

12. Lee JH*, Lee M*, **Lee H**, Park J, Kim S, Koyanagi A, Smith L, Kim MS, López Sánchez GF, Dragioti E, Rahmati M, Kang J[¶], Oh H, Yon DK[¶]. National trends in sexual intercourse and usage of contraception among Korean adolescents. *World J Pediatr*. 2024 Sep;20(9):935-948.
13. Kattih M*, **Lee H***, Jo H*, Jeong J*, Kim H*, Park J, Yang H, Nguyen A, Kim HJ, Lee H, Kim M, Lee M, Kwon R, Kim S, Koyanagi A, Kim MS, Rahmati M, López Sánchez GF, Dragioti E, Kim JH, Woo S, Cho SH[¶], Smith L[¶], Yon DK[¶]. National prevalence of atopic dermatitis in Korean adolescents from 2009 to 2022. *Sci Rep*. 2024 May 29;14(1):12391.

*Co-first author.

14. Kim H*, Son Y*, **Lee H***, Kang J, Hammoodi A, Choi Y, Kim HJ, Lee H, Fond G, Boyer L, Kwon R[¶], Woo S[¶], Yon DK[¶]. Machine Learning-Based Prediction of Suicidal Thinking in Adolescents by Derivation and Validation in 3 Independent Worldwide Cohorts: Algorithm Development and Validation Study. *J Med Internet Res*. 2024 May 17;26:e55913.

*Co-first author.

15. Hong S*, Woo S*, Kim S, Park J, Lee M, Kim S, Koyanagi A, Smith L, Kim MS, López Sánchez GF, Dragioti E, Rahmati M, Fond G, Boyer L, Oh J, **Lee H**[¶], Yon DK[¶]. National prevalence of smoking among adolescents at tobacco tax increase and COVID-19 pandemic in South Korea, 2005-2022. *Sci Rep*. 2024 Apr 3;14(1):7823.

[¶] Corresponding author

16. **Lee H***, Park J*, Lee M, Kim HJ, Kim M, Kwon R, Lee SW, Koyanagi A, Smith L, Kim MS, Rhee SY, Cho JK, Kim S, Rahmati M[¶], Yon DK[¶]. National Trends in Allergic Rhinitis and Chronic Rhinosinusitis and COVID-19 Pandemic-Related Factors in South Korea, from 1998 to 2021. *Int Arch Allergy Immunol*. 2024;185(4):355-361.

* Co-first author.

17. Lee JH*, Lee M*, **Lee H**, Park J, Woo S, Kim S, Koyanagi A, Smith L, Kim MS, López Sánchez GF, Dragioti E, Rahmati M, Fond G, Boyer L, Yon DK[¶]. National trends in sleep sufficiency and sleep time among adolescents, including the late-COVID-19 pandemic, 2009-2022: A nationally representative serial study in South Korea. *Asian J Psychiatr*. 2024 Mar; 93:103911.

18. **Lee H***, Cho JK*, Park J*, Lee H*, Fond G, Boyer L, Kim HJ, Park S, Cho W, Lee H, Lee J, Yon DK[¶]. Machine Learning-Based Prediction of Suicidality in Adolescents With Allergic Rhinitis: Derivation and Validation in 2 Independent Nationwide Cohorts. *J Med Internet Res*. 2024 Feb 14;26:e51473.

* Co-first author

19. Lee M*, **Lee H**, Park J, Kim HJ, Kwon R, Lee SW, Kim S, Koyanagi A, Smith L, Kim MS, Fond G, Boyer L, Rahmati M, Rhee SY[¶], Yon DK[¶]. Trends in hypertension prevalence, awareness, treatment, and

control in South Korea, 1998-2021: a nationally representative serial study. *Sci Rep*. 2023 Dec 8;13(1):21724.

20. Oh J*, Lee M*, **Lee H**, Yang H, Park J, Rahmati M, Koyanagi A, Smith L, Fond G, Boyer L, Kim MS, Lee SW, López Sánchez GF, Dragioti E, Woo HG, Yon DK[¶]. Hand and Oral Hygiene Practices of South Korean Adolescents Before and During the COVID-19 Pandemic. *JAMA Netw Open*. 2023 Dec 1;6(12):e2349249.
21. Hahn JW*, Yang HR, Moon JS, Chang JY, Lee K, Kim GA, Rahmati M, Koyanagi A, Smith L, Kim MS, López Sánchez GF, Elena D, Shin JY, Shin JI, Kwon R, Kim S, Kim HJ, **Lee H**, Ko JS[¶], Yon DK[¶]. Global incidence and prevalence of autoimmune hepatitis, 1970-2022: a systematic review and meta-analysis. *EClinicalMedicine*. 2023 Oct 17;65:102280.
22. Kang J*, Park J*, **Lee H**, Lee M, Kim S, Koyanagi A, Smith L, Kim MS, Rahmati M, Fond G, Boyer L, López Sánchez GF, Elena D, Cortese S, Kim T[¶], Yon DK[¶]. National trends in depression and suicide attempts and COVID-19 pandemic-related factors, 1998-2021: A nationwide study in South Korea. *Asian J Psychiatr*. 2023 Oct;88:103727.
23. Kim H*, Kwon R, **Lee H**, Lee SW, Rahmati M, Koyanagi A, Smith L, Kim MS, López Sánchez GF, Elena D, Yeo SG, Shin JI, Cho W[¶], Yon DK[¶]. Viral load dynamics and shedding kinetics of mpox infection: a systematic review and meta-analysis. *J Travel Med*. 2023 Sep 5;30(5):taad111.

HONORS/AWARDS/GRANTS/SCHOLARSHIPS

HONORS/AWARDS

- Best Paper Award, The 5th Engineering in Biomedical and Rehabilitation & Home Conference, Japan Jun. 2023

GRANTS/SCHOLARSHIPS

- AI Convergence Innovation Project Competition, Kyung Hee University **May. 2023 – Dec. 2023**
Thesis: Development of an Early Prediction and Personalized Health Management Platform for Chronic Diseases Using AI and Wearable Technology
- Academic Scholarship, Kyung Hee University **2024 Spring Semester**
- Academic Scholarship, Kyung Hee University **2023 Fall Semester**
- Academic Scholarship, Kyung Hee University **2023 Spring Semester**
- Awarded a fellowship by the Ministry of Science and ICT(IITP), South Korea **Dec. 2023 – Jun. 2024**

PROFESSIONAL EXPERIENCES

- **Researcher**, Asan Medical Center, Biomedical Engineering Research Center **Mar. 2025 – Feb. 2026**
Building **quantitative coronary angiography analysis pipelines** with **clinician QA interfaces**, improving measurement consistency and reviewer efficiency.

Prototyping **3D vascular geometry reconstruction from routine 2D angiographic sequences** for **interventional planning contexts**.

Developing **semi-automated instrument (catheter/wire) tracking** in angiographic videos using **foundation-model–assisted segmentation**.

Integrating **structured clinical variables** with model outputs to produce **clinician-facing explanations and summaries** for decision support.

Independent AI Research (DEFINES Project)

Seoul, South Korea | Feb 2026 – Present

- Extending prior LLM hallucination mitigation work (MIT/UofT, 2024) from text-based scientific Q&A to multimodal mathematical reasoning — investigating symbolic-neural hybrid architectures for constraining VLM outputs in structured knowledge domains.
- Designed 4-stage multi-agent pipeline: YOLOv8 document parsing → VLM formula extraction (Gemini 2.5 Flash via OpenRouter) → Neo4j knowledge graph alignment (Korean math curriculum ontology) → automated evaluation.
- Implemented knowledge constraint via PostgreSQL-Neo4j synchronization to suppress domain hallucination without fine-tuning.
- Established math-domain evaluation metrics (Normalized Levenshtein Distance, Character Error Rate) on ground-truth formula dataset.

- **M.S. in Pharmacy Researcher**, *Center for Digital Health Medical Science Research Institute*

Jan. 2023 – Feb. 2025

Developed and externally validated ML risk models for neurodegenerative, cardiovascular, and diabetic retinopathy outcomes in T2D across two **independent Korean cohorts** (JMIR 2024; Sci Rep 2024), standardizing the end-to-end pipeline (data curation → feature engineering → model benchmarking → **calibration**/external validation) with **clinician-interpretable** outputs.

Led **nationwide time-series analyses** of adolescent health behaviors in Korea (2005–2022/ 2009–2022), quantifying trends and policy impacts (tax hikes, COVID-19).

Contributed to population-level studies on allergic rhinitis/rhinosinusitis and hypertension (1998–2021), and to multi-country/meta-analytic projects (long COVID dermatologic events, mpox kinetics, autoimmune hepatitis).

Publications: 25 SCI/SCIE publications | 6 co-first author | 1 corresponding author | 404 citations | h-index 12 | Full list: scholar.google.com/citations?user=o2RGCNQAAAAJ&hl=en; established reproducible, code-first reporting (Python/R), automated tables/figures, and mentored junior analysts.

- **Internship**, *MEDIAZEN*

Jun. 2021 – Jan. 2022

Engineered text dataset preprocessing pipelines and generated structured training data (Python) for a commercial Korean-language chatbot system.

EXTRACURRICULAR ACTIVITIES

LAB TECHNIQUES

- Languages: Korean (native); English (professional working proficiency; **TOEFL iBT 94 (R24 L22 S23 W25, Nov. 2024)**)
- Data analysis of hospital data, survey data, and diabetic retina image dataset
- Factor analysis of cohort study

- Fine-tune pretrained LLMs; build small LLMs (sLLM) pipelines
- Programming: Python, R, MATLAB, C, JavaScript | SQL
- ML/DL: PyTorch, TensorFlow, scikit-learn | Interpretability (SHAP), calibration, time-series, survival/mixed models
- Imaging/Clinical: QCA (seg/centerline/bifurcation), SAM 2 for catheter tracking, clinical data modeling
- Tools: Git, Docker, Linux, Jupyter, SAS

ADDITIONAL EXPERIENCES

- CURT Program KUS University's Innovation Business Department
Prediction of Korea fine dust data using LSTM & GRU models **Apr. 2021 – Jul. 2022**
- Military Service in South Korea
Social Service Personnel at Seoul Metro **Nov. 2018 – Sep. 2020**
Certificate of Commendation
- Family Education Team Mentoring Service Volunteer
Gangnam-gu Family Support Center (113 hours) **Jul. 2017 – Sep. 2018**